Exercise 1

Happy Supplies
Parts Warehouse

Customer Name:	Jeff Peterson	**Date:**	7/1/2024
Customer Number:	HG54587	**Time:**	10:30am
		Employee:	D. Harrison
Customer Type:	Consumer		

Part Number	Name	Type	Cage Code	Quantity Ordered	Unit Price
10654	Float Control	Plumbing	G413	4	12
10456	Modulator	Electrical	H433	3	7
10776	Hose Assembly	Plumbing	G413	7	9
10657	Float Assembly	Plumbing	G413	5	10

Assumptions:

1. Employees help any and all different customers (no special relationship between specific employees and customers)
2. Each customer has a unique customerNumber, and each part has a unique partNumber.
3. A cageCode is the identifier of the cages/shelves that the inventory is stored in, and multiple parts that belong to the same type can be stored in the same cage/shelf (share the same cage code).
4. Every customer can place many orders, and every order can conain multiple parts.
5. The same part can belong to multiple orders.
6. Unit price is specific to each part.

customerNumber (PK)	customerName	customerType	date (PK)	time (PK)	employee	partNumber (PK)	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

Normalizing Process

The above table is already in 1NF because there's primary key (composite PK, customerNumber+date+time+partNumber), and there's no repeating groups.

1NF →2NF:

Problem: There are partial dependencies.
customerName, customerType are only dependent on customerNumber.
employee is only dependent on (customerNumber, date, time).
partName, partType, cageCode, unitPrice are only dependent on partNumber.
So, we separate them into four tables as below:

customer

customerNumber (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer

order

customerNumber (PK)	date (PK)	time (PK)	employee
HG54587	7/1/2024	10:30am	D. Harrison

part

partNumber (PK)	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_part

customerNumber (PK)	date (PK)	time (PK)	partNumber (PK)	quantityOrdered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5

2NF →3NF:

Problem: There is transitive dependencies.
A transitive dependency exists here because cageCode depends on partType, not directly on partNumber.
So, we separate **part** into 2 tables as below:

customer

customerNumber (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer

order

customerNumber (PK, FK)	date (PK)	time (PK)	employee
HG54587	7/1/2024	10:30am	D. Harrison

part

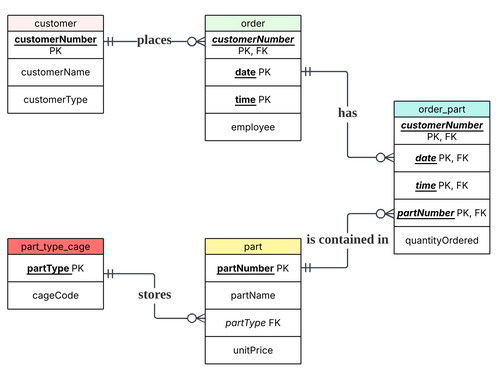
partNumber (PK)	partName	partType (FK)	unitPrice
10654	Float Control	Plumbing	12
10456	Modulator	Electrical	7
10776	Hose Assembly	Plumbing	9
10657	Float Assembly	Plumbing	10

part_type_cage

partType (PK)	cageCode
Plumbing	G413
Electrical	H433

order_part

customerNumber (PK, FK)	date (PK, FK)	time (PK, FK)	partNumber (PK, FK)	quantityOrdered
HG54587	7/1/2024	10:30am	10654	4
HG54587	7/1/2024	10:30am	10456	3
HG54587	7/1/2024	10:30am	10776	7
HG54587	7/1/2024	10:30am	10657	5



Exercise 2

staffNo	therapistName	patNo	patName	appointment date	appointment time	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Assumptions:

1. Therapists may work at a number of different branches, but they only see patients at one specific branch on any given day (appointment date), and they can only see one patient at any given appointment time on any appointment date.
2. A patient is given an appointment at a specific time and date at a particular branch with one therapist.
3. Patients may have multiple appointments in any given day and with multiple different therapists.
4. Each therapist has a unique staffNo, and each patient has a unique patNo.
5. A branch is uniquely defined by staffNo and appointmentDate, and each branch can have multiple therapists per day
6. An appointment is uniquely defined by staffNo, appointmentDate, appointmentTime.

staffNo (PK)	therapistName	patNo (PK)	patName	appointmentDate (PK)	appointmentTime (PK)	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy Mckee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy Mckee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Normalizing Process

The above table is already in 1NF because there's primary key (composite PK, staffNo+patNo+appointmentDate+appointmentTime), and there's no repeating groups.

1NF →2NF:

Problem: There are partial dependencies.
therapistName is only dependent on staffNo.
patName is only dependent on patNo.
branchNo is only dependent on staffNo and appointmentDate.
So, we separate them into four tables as below:

therapist

staffNo (PK)	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient

patNo (PK)	patName
P100	Lily White
P105	Jill Baker
P108	Andy Mckee
P110	Jimmy Winter

branch

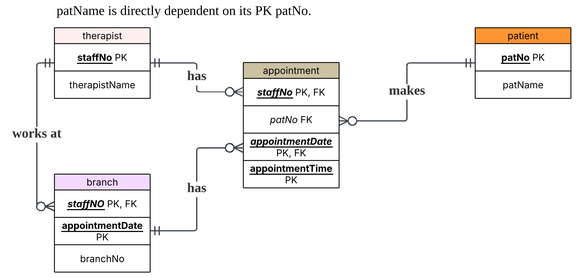
staffNo (PK, FK)	appointmentDate (PK)	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

staffNo (PK, FK)	patNo FK	appointmentDate (PK, FK)	appointmentTime (PK)
S1011	P100	9/12/2022	10:00
S1011	P105	9/12/2022	12:00
S1024	P108	9/12/2022	10:00
S1024	P108	9/14/2022	14:00
S1032	P105	9/14/2022	16:30
S1032	P110	9/15/2022	18:00

2NF →3NF:

The tables above are already in 3NF because there's no transitive dependencies.
branchNo is directly dependent on its composite PK (staffNo, appointmentDate)
therapistName is directly dependent on its PK staffNo.
patName is directly dependent on its PK patNo.



Exercise 3

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Assumptions:

1. eNo is unique for each member of staff.
2. contractNo is unique for each contract.
3. eventNo is unique for each event.
4. Each contract only applies to one event, and can be handled by multiple staff members.
5. An event might have different contracts.
6. A staff member can work on multiple contracts.

eNo (PK)	contractNo (PK)	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Normalizing Process

The above table is already in 1NF because there's primary key (composite PK, eNo+contractNo), and there's no repeating groups.

1NF →2NF:

Problem: There are partial dependencies.
eName is only dependent on eNo.
eventNo is only dependent on contractNo.
So, we separate them into three tables as below:

employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

contractNo (PK)	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

employee_contract_assignment

eNo (PK, FK)	contractNo (PK, FK)	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

2NF →3NF:

Problem: There is transitive dependencies.
A transitive dependency exists here because eventLoc depends on eventNo, not directly on contractNo.
So, we separate **contract** into 2 tables as below. After separating, it becomes 3NF since there's no more transitive dependencies.

employee

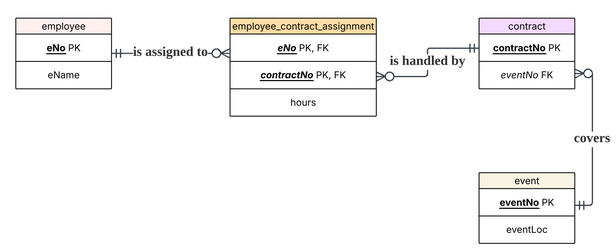
eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

contractNo (PK)	eventNo (FK)
C1024	H25
C1025	H4
C1026	H25

employee_contract_assignment

eNo (PK, FK)	contractNo (PK, FK)	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

event

eventNo (PK)	eventLoc
H25	Queens
H4	Yonkers